MA1220 **- Calculus II** Solution

Date: October 14, 2023 Maximum Marks 50

Time: 120 minutes, 09:30 AM - 11:30 AM

Q.1 Answer the following. (No justification required). (**Note that you must answer Q.1 on page number 2 only.**) (8 marks)

(a) If $f(x) \geq g(x)$ on the interval $[a, \infty)$ then, if $\int_a^\infty f(x)dx$ converges then $\int_a^\infty g(x)dx$ also converges. **Answer: False**

(b) $\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial^2 f}{\partial y \partial x}$ is always true. **Answer: False**

(c) If $f_x = f_y = 0$ at a point (x, y) then f is differentiable at (x, y). **Answer: False**

(d) $D_{\mathbf{v}}\left(\dfrac{\partial f}{\partial x_2}\right)(-3, 4) = \dfrac{\partial^2 f}{\partial x_1 \partial x_2}(-3, 4)$ if $\mathbf{v} = \langle 1, 0\rangle$ **Answer: True**

(e) If $f(1) = 12$, f' is continuous, and $\int_1^4 f'(x)\, dx = 17$, then $f(4) =$?. **Answer:** $f(4) = 29$**.**

(f) Is $\int_{-1}^1 \frac{1}{x}\, dx$ convergent or divergent? **Answer: Divergent**

(g) Describe the domain $\mathcal{D}$ of the function $f(x, y) = \sqrt{1 - x^2 - y^2} \cdot \ln\left(x^2 + y^2 - 1\right)$.
Answer: $\mathcal{D} = \{(x, y) \in \mathbb{R}^2 \mid 1 - x^2 - y^2 \geq 0 \text{ and } x^2 + y^2 - 1 > 0\} = \emptyset$.

(h) Is it true that $\int_{-\infty}^\infty x\, dx = \lim_{t\to\infty} \int_{-t}^t x\, dx$. **Answer: False**

Solution (h) and (f). Check that for any $t \in \mathbb{R}$, one has $\int_{-t}^t x\, dx = 0$. Then $\lim_{t\to\infty} 0 = 0$. The L.H.S. integral using improper integral definition can be written in the form

$$\int_{-\infty}^c x\, dx + \int_c^\infty x\, dx$$

for some constant c. Notice that each of these integrals is divergent. Hence, the sum is divergent. The solution for (f) is exactly the same.

Q.2 Answer the following:

(a) Find the directional derivative of the function $f(u, v) = u^2 e^{-v}$, at the point $(3, 0)$ in the direction of the vector $3\,\mathbf{i} + 4\,\mathbf{j}$. (3 marks)

(b) By comparing (without integrating,) determine whether the integral $\int_1^\infty \dfrac{1}{\sqrt{x^3 + 1}}\, dx$ converges or diverges. (2 marks)

(c) Find the local minima, maxima, saddle points of the function $f(x, y) = x^2 + xy + y^2 + y$ in $\mathbb{R}^2$. (3 marks)

Solution of (a). The directional derivative of a function $f(u, v)$ in the direction of a unit vector $\mathbf{w} = a\mathbf{i} + b\mathbf{j}$ is given by

$$D_{\mathbf{w}} f(u, v) = \nabla f(u, v) \cdot \mathbf{w} = \frac{\partial f}{\partial u} a + \frac{\partial f}{\partial v} b$$

Given $f(u,v) = u^2 e^{-v}$, we have:

$$\frac{\partial f}{\partial u} = 2ue^{-v} \quad \text{and} \quad \frac{\partial f}{\partial v} = -u^2 e^{-v} \qquad \text{(1 mark)}$$

The unit vector $\mathbf{w}$ in the direction of $3\,\mathbf{i} + 4\,\mathbf{j}$ is given by

$$\mathbf{w} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}. \qquad \text{(1 mark)}$$

Substituting the partial derivatives and the values of u and v, we get:

$$D_{\mathbf{w}} f(3,0) = \frac{3}{5} f_u(3,0) + \frac{4}{5} f_v(3,0) = \frac{3}{5} \cdot 6 - \frac{4}{5} \cdot 9 = \frac{18-36}{5} = -\frac{18}{5}. \qquad \text{(1 mark)}$$

Common mistakes. Many have not written the unit vector $\mathbf{w} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$ for computing directional derivatives. Other mistake is writing the directional derivative $D_{\mathbf{w}} f(3,0) = \dfrac{18}{5}\mathbf{i} - \dfrac{36}{5}\mathbf{j}$ as vector which is wrong.

Solution of (b). Given $x > 1$ (this assumption is important), one has

$$\sqrt{1+x^3} \geq x^{\frac{3}{2}} \Rightarrow \frac{1}{\sqrt{1+x^3}} \leq \frac{1}{x^{\frac{3}{2}}} \qquad \text{(1 mark)}$$

We had shown in class that $\int_1^\infty x^{-p}$ converges when $p > 1$. In our case, $p = \frac{3}{2}$; hence, by comparison test, the integral converges. (1 mark)

Common mistakes. Many have taken the wrong comparison function. Another mistake is to calculate the integral explicitly. The hypothesis $x > 1$ is important in comparing two functions.

Solution of (c). Given $f(x,y) = x^2 + xy + y^2 + y$. Compute the partial derivatives, we get $\frac{\partial f}{\partial x} = 2x + y$ and $\frac{\partial f}{\partial y} = x + 2y + 1$. Solve the equations

$$2x + y = 0$$
$$x + 2y + 1 = 0$$

to get the critical point $(\frac{1}{3}, -\frac{2}{3})$. (1 mark)

The second partial derivatives are $\frac{\partial^2 f}{\partial x^2} = 2$, $\frac{\partial^2 f}{\partial y^2} = 2$ and $\frac{\partial^2 f}{\partial x \partial y} = 1$. The discriminant D for the second partial derivative test is

$$D = \frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 = 2 \cdot 2 - 1 = 3 > 0. \qquad \text{(1 mark)}$$

Since $D > 0$ and $\frac{\partial^2 f}{\partial x^2} = 2 > 0$, hence the critical point is a local minimum. (1 mark)

Common mistakes. The expression for the partial derivatives is wrong.

Q.3 Answer the following:

(a) Find the volume V obtained by rotating the region enclosed by the graphs of the functions $f(x) = x(5-x)$ and $g(x) = 8 - x(5-x)$ about the y-axis. (4 marks)

(b) Find the area of the region bounded by the graphs of $y = \dfrac{8}{x^2}$, $y = 8x$, and $y = x$. (4 marks)

Solution of (a). From the equation $f(x) = g(x)$, that is, when $x(5-x) = 8 - x(5-x)$, we get $x = 1$ and $x = 4$. (1 mark)

Notice that $f(x) \geq g(x)$ for all $x \in [1,4]$. With $f(x) \geq g(x)$ for all $x \in [1,4]$, the required volume is

$$V = \int_1^4 2\pi x(f(x) - g(x))\, dx. \qquad \text{(1 mark)}$$

Then

$$V = \int_1^4 2\pi x(10x - 2x^2 - 8)\, dx \qquad \text{(1 mark)}$$

Calculating the integral, we get ,

$$V = 45\pi. \qquad \text{(1 mark)}$$

Common mistakes. Many of you have written the wrong formula for computing the volume. One such mistake is

$$V = \int_1^4 \pi \left(x_f^2(y) - x_g^2(y)\right)\, dx.$$

See question, rotation is about the y-axis.

Solution of (b). From $\dfrac{8}{x^2} = 8x$, we get $x^3 = 1$, and hence $x = 1$. From $\dfrac{8}{x^2} = x$, we get $x^3 = 8$, and hence $x = 2$. (1 mark)

Observe that for all $x \in [0,1]$, one has $8x > x$ and for all $x \in [1,2]$, one has $\dfrac{8}{x^2} > x$. Hence, the required area is

$$A = A_1 + A_2,$$

where

$$A_1 = \int_0^1 (8x - x)\, dx, \qquad \text{(1 mark)}$$

and

$$A_2 = \int_1^2 (\frac{8}{x^2} - x)\, dx. \qquad \text{(1 mark)}$$

Computing the integrals, we get

$$A = 6 \qquad \text{(1 mark)}$$

Q.4 Answer the following:

(a) If the region $\mathcal{R} = \{(x,y) \mid x \geq 1,\ 0 \leq y \leq \frac{1}{x}\}$ is rotated about the x-axis, by using improper integral show that:

(i) The volume of the resulting solid is finite, and find the volume. (2 marks)

(ii) The surface area is infinite. (3 marks)

(b) If $f(x,y) = \dfrac{x^{2023}\, y}{x^{4046} + y^2}$, does (3 marks)

$$\lim_{(x,y)\to(0,0)} f(x,y) \text{ exist?}$$

Solution of (a). Part (i). The volume of solid obtained by rotation of $y = \frac{1}{x}$ about x-axis for $x = 1$ to some fixed point $b > 1$ is given by

$$\int_1^b \pi(\frac{1}{x})^2\, dx. \qquad \text{(1 mark)}$$

Calculate

$$\pi \int_1^b \left(\frac{1}{x}\right)^2 dx = -\pi \frac{1}{x}]_1^b = \pi(1 - \frac{1}{b}).$$

and hence

$$\lim_{b\to\infty} \pi \int_1^b \left(\frac{1}{x}\right)^2 dx = \lim_{b\to\infty} \pi(1 - \frac{1}{b}) = \pi. \qquad \text{(1 mark)}$$

The volume is finite.

Part (ii). The surface area of solid by rotation of $y = \frac{1}{x}$ about x-axis for $x = 1$ to some fixed point $b > 1$ is given by

$$S = \int_1^b 2\pi \frac{1}{x} \sqrt{1 + (\frac{dy}{dx})^2} dx. \qquad \text{(1 mark)}$$

Since

$$\sqrt{1 + (\frac{dy}{dx})^2} = \sqrt{1 + \frac{1}{x^4}} > 1, \quad \text{for all } x \geq 1,$$

therefore

$$\frac{1}{x}\sqrt{1 + (\frac{dy}{dx})^2} > \frac{1}{x} \quad \text{for all } x \geq 1. \qquad \text{(1 mark)}$$

Thus, by comparison test, the integral $\int_1^\infty 2\pi \frac{1}{x} \sqrt{1 + (\frac{dy}{dx})^2} dx$ diverges. This proves that the surface area is infinite. (1 mark)

Common mistakes. Many of you have directly written that integral diverges and surface area is infinite, which is incorrect. You must show that the integral diverges either by a comparison test or by any other method.

Solution of (b). Let $x = my$ be any line through the origin $(0, 0)$, then

$$f(x, y) = f(my, y) = \frac{m^{2023}\, y^{2023} y}{m^{4046}\, y^{4046} + y^2} = \frac{m^{2023}\, y^{2022}}{m^{4046} y^{4044} + 1}.$$

$$\text{So} \quad f(x, y) \to 0 \quad \text{as} \quad (x, y) \to (0, 0) \text{ along } x = my. \qquad \text{(1 mark)}$$

Now let $(x, y) \to (0, 0)$ along the curve $y = x^{2023}$. We get

$$f(x, y) = f\left(x, x^{2023}\right) = \frac{x^{2023}\, x^{2023}}{x^{4046} + x^{4046}} = \frac{x^{4046}}{2\, x^{4046}} = \frac{1}{2}.$$

so

$$f(x, y) \to \frac{1}{2} \quad \text{as} \quad (x, y) \to (0, 0) \text{ along the curve } y = x^{2023}. \qquad \text{(1 mark)}$$

Since different paths lead to different limiting values, the given limit does not exist. (1 mark)

Q.5 Answer the following:

(a) Determine the function $f(x)$ and all values of a such that (2 marks)

$$\int_a^x f(t)\, dt = x^2 + x - 6.$$

(b) Evaluate $g''(\frac{\pi}{6})$, where (2 marks)

$$g(y) = \int_3^y f(x)\, dx \quad \text{and} \quad f(x) = \int_0^{\sin x} \sqrt{1 + t^2}\, dt.$$

(c) Using Riemann integration, find the value of (2 marks)

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2+k^2}.$$

(d) Does there exists a function $f(x,y)$ such that $f_x(x,y)=xy$ and $f_y(x,y)=x^2$? If yes, find f. If no, explain your reason. (2 marks)

Solution of (a). Applying FTC Part II, we get

$$f(x)=\frac{d}{dx}(x^2+x-6)=2x+1. \qquad \text{(1 mark)}$$

Therefore,

$$\int_a^x (2t+1)\,dt = x^2+x-(a^2+a).$$

Then $a^2+a=6$. This implies $a=2$ or $a=-3$. (1 mark)

Solution of (b). Since $g(y)=\int_3^y f(x)\,dx$, we can write

$$g'(y)=f(y) \qquad \text{(1 mark)}$$

and since $f(x)=\int_0^{\sin x}\sqrt{1+t^2}\,dt$, we can write

$$g''(y)=f'(y)=\sqrt{1+\sin^2 y}\cdot\cos y.$$

Thus

$$g''(\frac{\pi}{6})=\sqrt{1+\left(\frac{1}{2}\right)^2}\cdot\frac{\sqrt{3}}{2}=\frac{\sqrt{15}}{4}. \qquad \text{(1 mark)}$$

Solution of (c).

(1)
$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2+k^2}=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{1}{1+(\frac{k}{n})^2}.$$

This is the Riemann right sum for

$$\int_0^1 \frac{1}{1+x^2}dx. \qquad \text{(1 mark)}$$

Hence

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2+k^2}=\int_0^1\frac{1}{1+x^2}dx=\left[\tan^{-1}x\right]_0^1=\frac{\pi}{4}. \qquad \text{(1 mark)}$$

Solution of (d). Show that f cannot satisfy Clairaut's Theorem; hence, no such function exists. (2 marks)

Common mistakes. Most of you have converted the given problems into another equivalent problem. Your approach goes like this: Integrating $f_x(x,y)=xy$ w.r.t. x, we get

$$f(x,y)=\frac{x^2y}{2}+h_1(y).$$

Similarly, integrating $f_y(x,y)=x^2$ w.r.t. y, we get

$$f(x,y)=x^2y+h_2(x).$$

From these equations, we can write

$$\frac{x^2y}{2} + h_1(y) = x^2y + h_2(x)$$

that is,

$$h_1(y) - h_2(x) = \frac{x^2y}{2}.$$

Since no such $h_1(y)$ and $h_2(x)$ exist, there is no solution to the original question. The equivalent question could have been, prove that there are no functions $h_1(y)$ and $h_2(x)$ for which $h_1(y) - h_2(x) = \frac{x^2y}{2}$. You are given a partial mark for this approach (1 mark).

Q.6 Answer the following:

(a) Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be a function defined by (4 marks)

$$f(x, y) = \sigma x + \pi y \quad \text{for all } (x, y) \in \mathbb{R}^2$$

where σ and π are constants. Prove (**using $\epsilon - \delta$ definition**) that f is continuous on $\mathbb{R}^2$.

(b) Let $g : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be a function defined by $g(x, y) = 5x + 4y^2$. Prove (**using the steps given below**) that $g(x, y) = 5x + 4y^2$ is differentiable at $(2, 1)$. (4 marks)

(Step 1.) Compute linearization function $L(x, y)$ (the equation of tangent plane) of $g(x, y)$ at $(2, 1)$.

(Step 2.) Compute the difference $g(x, y) - L(x, y)$.

(Step 3.) Show that for all $(x, y) \neq (2, 1)$, one can write

$$0 \leq \frac{f(x, y) - L(x, y)}{\sqrt{(x-2)^2 + (y-1)^2}} \leq A|y - 1|$$

for some constant A. Find one such A.

(Step 4.) Apply the definition of differentiability for $g(x, y)$ at $(2, 1)$.

Solution of (a). Let $(a, b) \in \mathbb{R}^2$ be any arbitrary point. We will show that f is continuous at (a, b).

Let $\varepsilon > 0$ be an arbitrary small positive number. We want to find $\delta > 0$ such that if

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \quad \text{then } |f(x, y) - f(a, b)| < \varepsilon.$$

Equivalently, if

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \quad \text{then} \quad |\sigma(x-a) + \pi(y-b)| < \varepsilon.$$

Note that

$$|x - a| \leq \sqrt{(x-a)^2 + (y-b)^2} \quad \text{and} \quad |y - b| \leq \sqrt{(x-a)^2 + (y-b)^2}. \qquad \text{(1 mark)}$$

Now we can write

$$\begin{aligned} |f(x, y) - f(a, b)| &= |\sigma(x-a) + \pi(y-b)| \\ &\leq |\sigma|\,|x-a| + |\pi|\,|y-b|\,. \qquad \text{(1 mark)} \end{aligned}$$

Let $M := \max\{|\sigma|, |\pi|\}$, then

$$\begin{aligned} |f(x, y) - f(a, b)| &\leq |\sigma|\,|x-a| + |\pi|\,|y-b| \\ &\leq M\,|x-a| + M\,|y-b| \\ &\leq 2M\sqrt{(x-a)^2 + (y-b)^2}. \qquad \text{(1 mark)} \end{aligned}$$

Now, we have to find a $\delta > 0$ such that

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \text{ then } |f(x,y) - f(a,b)| < \varepsilon.$$

If $M = 0$, then the function $f(x,y) = 0$ for all $(x,y) \in \mathbb{R}^2$, which is of course a continuous function. So, without loss of generality, we can assume $M > 0$. Thus any given $\varepsilon > 0$ if we choose $\delta = \frac{\varepsilon}{2M}$ then

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \Rightarrow |f(x,y) - f(a,b)| < \varepsilon. \qquad \text{(1 mark)}$$

Hence, by definition,

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b),$$

i.e. $f(x,y)$ is continuous at (a,b). Since (a,b) is an arbitrary point $f(x,y)$ is continuous on $\mathbb{R}^2$.

Common mistakes. Many of you have proven continuity at $(0,0)$ only and concluded that that implies continuity at all points (a,b) on $\mathbb{R}^2$, which is incorrect. You need to prove at an arbitrary point (a,b) on $\mathbb{R}^2$ and then conclude that proof holds for all points of $\mathbb{R}^2$.

Solution of (b). The linearization function $L(x,y)$ (the equation of tangent plane) of $g(x,y)$ at $(2,1)$ is

$$L(x,y) = g(2,1) + g_x(2,1)(x-2) + g_y(2,1)(y-1).$$

Since $g_x = 5$ and $g_y = 8y$, thus $g_x(2,1) = 5$ and $g_y(2,1) = 8$. Therefore

$$L(x,y) = 14 + 5(x-2) + 8(y-1). \qquad \text{(1 mark)}$$

Simplifying the expression, we get

$$L(x,y) = 5x + 8y - 4.$$

Now,

$$g(x,y) - L(x,y) = (5x + 4y^2) - (5x + 8y - 4) = 4y^2 - 8y + 4 = 4(y-1)^2.$$

For all $(x,y) \neq (2,1)$, one can write

$$\frac{g(x,y) - L(x,y)}{\sqrt{(x-2)^2 + (y-1)^2}} = \frac{4(y-1)^2}{\sqrt{(x-2)^2 + (y-1)^2}}.$$

Since $(y-1)^2 \leq (x-2)^2 + (y-1)^2 \implies \frac{1}{\sqrt{(x-2)^2+(y-1)^2}} \leq \frac{1}{\sqrt{(y-1)^2}}$, we can write the following

$$0 \leq \frac{4(y-1)^2}{\sqrt{(x-2)^2 + (y-1)^2}} \leq 4|y-1| \qquad \text{(2 marks)}$$

One can take A as any constant real number ≥ 4.
When

$$\lim_{(x,y)\to(2,1)} 4|y-1| = \lim_{y\to 1} = 4|y-1| = 0.$$

Thus, we get

$$\lim_{(x,y)\to(2,1)} \frac{g(x,y) - L(x,y)}{\sqrt{(x-2)^2 + (y-1)^2}} = 0. \qquad \text{(1 mark)}$$

We can conclude now that the given function $g(x,y) = 5x + 4y^2$ is differentiable at $(2,1)$.